Lecture 7: Linear Transformation and Projection

MATH203 Applied Linear Algebra

From Static to Dynamic

Recap: What We Know So Far

In Lectures 1–6, matrices were **static objects**:

- Ingredient tables (recipes for making products)
- Containers of column vectors and row vectors
- Objects with four fundamental subspaces

Today: Matrices become **dynamic** — they **act** on vectors.

$$\mathbf{v} \xrightarrow{A} A\mathbf{v}$$

Roadmap for Today

Part 1: Linear Transformations

- What does "preserving linear structure" mean?

Part 2: Projection Operators ($P^2 = P$)

- The sunlight-and-floor model

Part 3: Basic Properties of Projections

- Interchanging property, disjoint property

Part 4: Orthogonal Projections ($P = P^T$)

- When sunlight is perpendicular to the floor

Linear Transformations

Definition

Definition (Linear Transformation)

A function $T : V \to W$ between vector spaces is a **linear transformation** if

$$T(\lambda \mathbf{v} + \mu \mathbf{w}) = \lambda\, T(\mathbf{v}) + \mu\, T(\mathbf{w})$$

for all $\mathbf{v}, \mathbf{w} \in V$ and all scalars λ, μ.

Equivalently, two separate properties:

1. **Additivity**: $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$
2. **Homogeneity**: $T(\lambda \mathbf{v}) = \lambda\, T(\mathbf{v})$

Linearity Extends to Any Linear Combination

Proposition (General Linearity)

If T is a linear transformation, then for any vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ and scalars $\lambda_1, \ldots, \lambda_k$:

$$T(\lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k) = \lambda_1 T(\mathbf{v}_1) + \cdots + \lambda_k T(\mathbf{v}_k)$$

Proof (induction on k): Base case $k = 2$ is the definition.

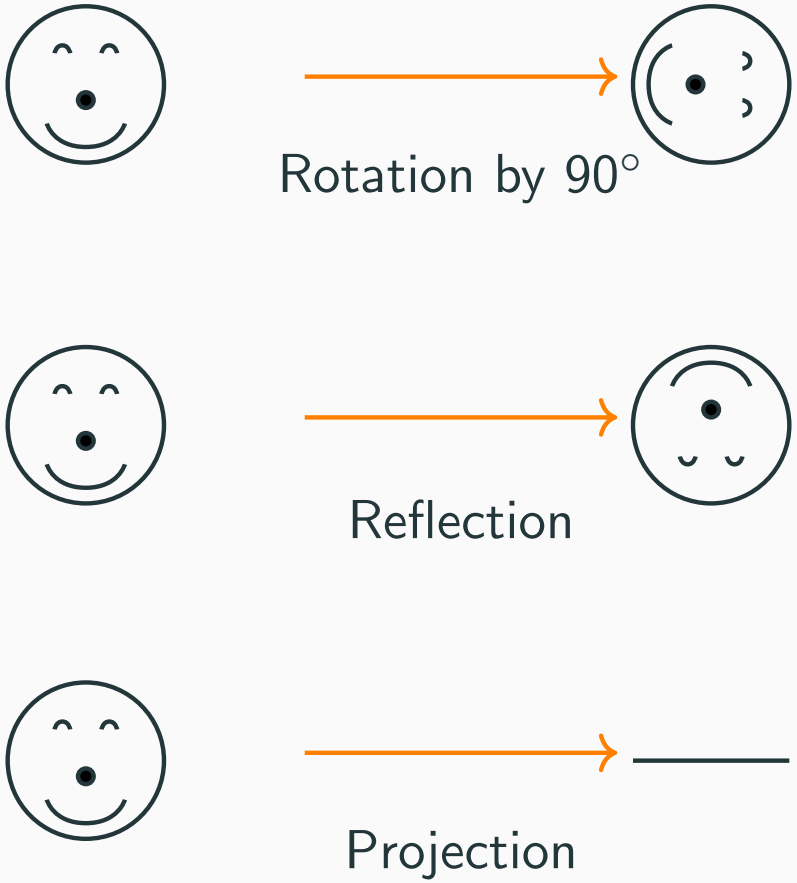
Inductive step: let $\mathbf{u} = \lambda_1 \mathbf{v}_1 + \cdots + \lambda_{k-1} \mathbf{v}_{k-1}$. Then:

$$\begin{aligned} T(\mathbf{u} + \lambda_k \mathbf{v}_k) &= T(\mathbf{u}) + \lambda_k T(\mathbf{v}_k) && \text{(definition)} \\ &= \lambda_1 T(\mathbf{v}_1) + \cdots + \lambda_k T(\mathbf{v}_k) && \text{(induction hypothesis)} \end{aligned}$$

$\square$

Geometric Meaning

Linear transformations **preserve linear structure**. Geometrically, a linear transformation preserves all parallel lines — it keeps the **shape** of a grid, but not necessarily size or angles:



Common Examples in $\mathbb{R}^2$

1. **Rotation by 90°**: $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix} \quad A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$
2. **Reflection across x-axis**: $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix} \quad A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Common Examples in $\mathbb{R}^2$ (continued)

3. **Scaling by 2**: $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix} \quad A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

4. **Projection onto x-axis**: $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix} \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

In each case: the matrix A **is** the linear transformation.

Matrix Representation

Proposition (Matrix Representation)

Every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be written as

$$T(\mathbf{v}) = A\mathbf{v}$$

for some $m \times n$ matrix A.

Conversely, every matrix defines a linear transformation.

Definition (Linear Operator)

When the domain and codomain are the same ($T : V \to V$), we call T a **linear operator**.

Represented by **square matrices**.

For the rest of this lecture, we focus on **linear operators** (square matrices).

Projection Operators

Definition

Definition (Projection Operator (Idempotent))

A matrix P is called a **projection operator** if

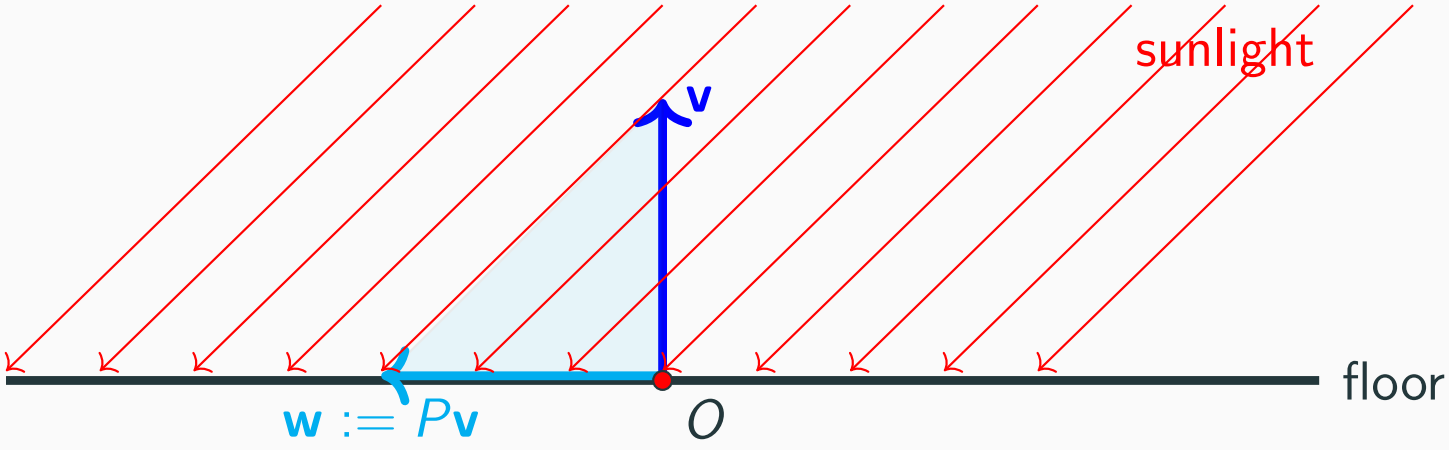
$$P^2 = P$$

The name "idempotent" comes from Latin:

$$\textit{idem}\ (\text{same}) + \textit{potent}\ (\text{power})$$

Applying the operation multiple times gives the **same result** as applying it once.

The Sunlight-and-Floor Model



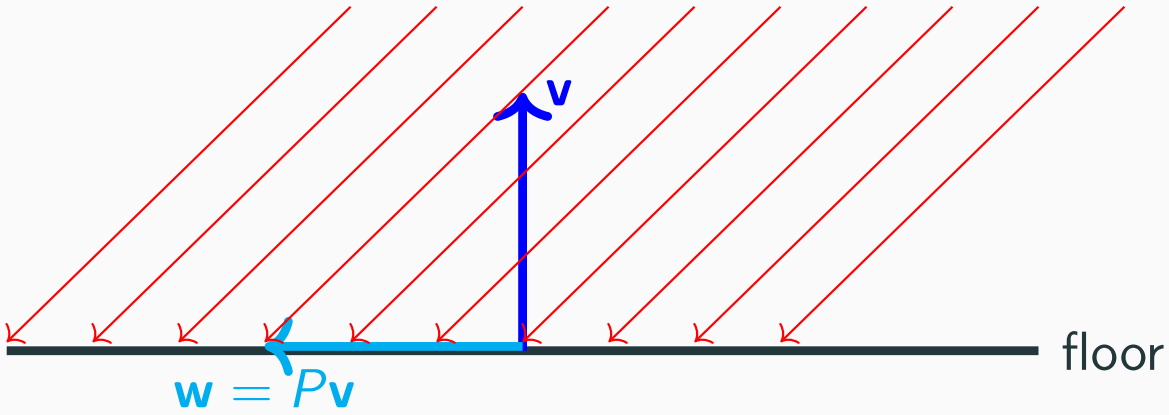
- A **vector v** stands at the origin (like a tree)
- **Sunlight** shines from some direction
- The projection P maps $\mathbf{v}$ to its **shadow** $\mathbf{w} = P\mathbf{v}$ on the floor

The sunlight direction **need not** be perpendicular to the floor!

Why $P^2 = P$?

We require $P^2 = P$. This means $P^2\mathbf{v} = P\mathbf{v}$ for any $\mathbf{v}$.

Geometrically: **The shadow of a shadow is the shadow itself.**



Since $\mathbf{w} = P\mathbf{v}$ lies on the floor, the shadow of $\mathbf{w}$ is **itself**:

$$P\mathbf{w} = \mathbf{w} \quad \Longrightarrow \quad P(P\mathbf{v}) = P\mathbf{v} \quad \Longrightarrow \quad P^2 = P$$

Three Key Questions

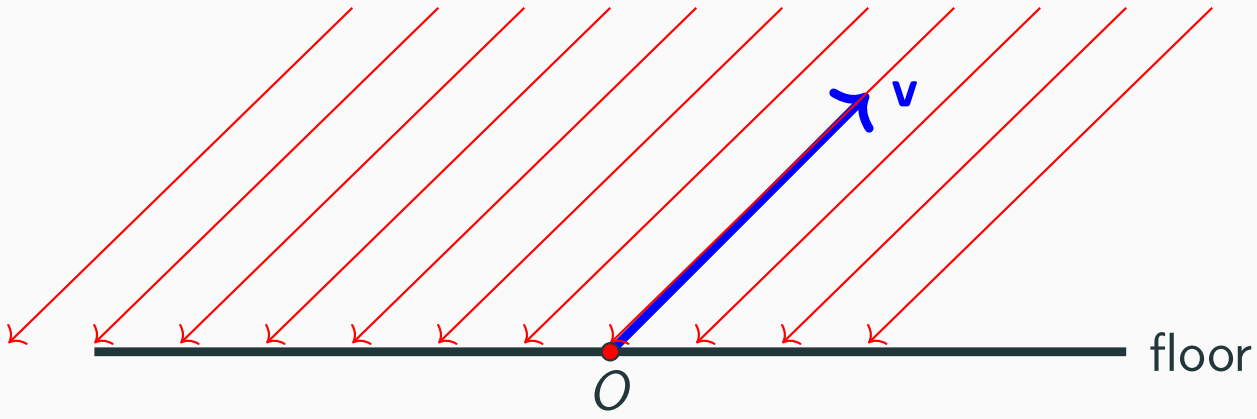
We defined projection only by $P^2 = P$. But intuitively a projection needs:

1. If $P^2 = P$, how do we define **the floor**?
2. If $P^2 = P$, how do we define **the direction of sunlight**?
3. After specifying these, why is $P\mathbf{v}$ exactly **the shadow** of $\mathbf{v}$?

Let us answer them one by one.

Question 1: Vectors Pointing Toward the Sun

If a vector $\mathbf{v}$ points **toward the sun**, what is its shadow?



Answer: $P\mathbf{v} = \mathbf{0}$ (no shadow!)

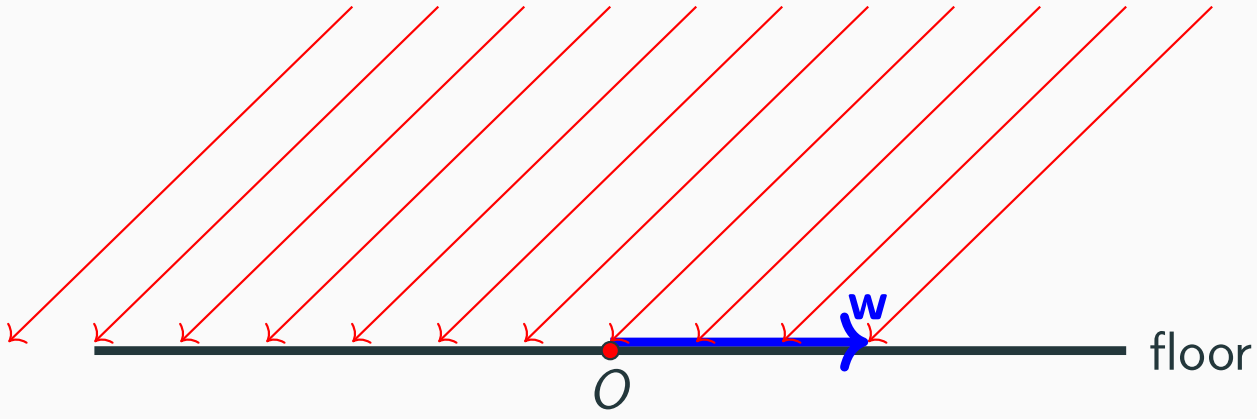
The set of all such vectors:

$$\{\mathbf{v} \in V : P\mathbf{v} = \mathbf{0}\} = \text{Null}(P)$$

$\text{Null}(P)$ describes the **direction of sunlight**.

Question 2: Vectors Lying on the Floor

If a vector $\mathbf{w}$ lies **on the floor**, what is its shadow?



Answer: $P\mathbf{w} = \mathbf{w}$ (a vector on the floor is its own shadow)

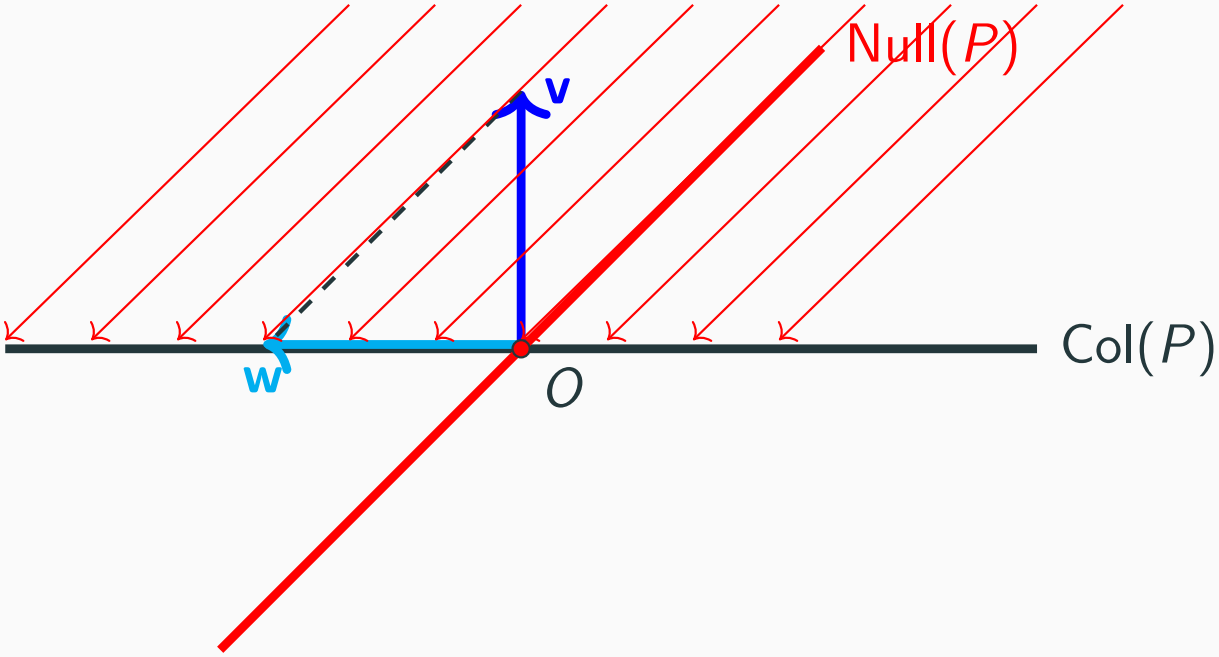
Two equivalent descriptions of the floor:

$$\underbrace{\{\mathbf{w} : \mathbf{w} = P\mathbf{v} \text{ for some } \mathbf{v}\}}_{\mathrm{Col}(P)} = \underbrace{\{\mathbf{w} : P\mathbf{w} = \mathbf{w}\}}_{\text{fixed points of } P}$$

$\mathrm{Col}(P)$ describes **the floor**.

Question 3: Why Is $P\mathbf{v}$ the Shadow?

Given $P^2 = P$, with the floor $\text{Col}(P)$ and sunlight $\text{Null}(P)$:



The shadow $\mathbf{w}$ must satisfy:

1. $\mathbf{w} \in \text{Col}(P)$ (shadow lies on the floor)
2. $\mathbf{v} - \mathbf{w} \in \text{Null}(P)$ (line from $\mathbf{v}$ to $\mathbf{w}$ follows sunlight)

The Proof

Proposition (Projection gives the shadow)

If $P^2 = P$ and $\mathbf{w}$ is the geometric shadow of $\mathbf{v}$ (i.e. $\mathbf{w} \in \mathrm{Col}(P)$ and $\mathbf{v} - \mathbf{w} \in \mathrm{Null}(P)$), then $\mathbf{w} = P\mathbf{v}$.

Proof:

Since $\mathbf{w} \in \mathrm{Col}(P)$, we have $\mathbf{w} = P\mathbf{x}$ for some $\mathbf{x}$.

$$P\mathbf{w} = P(P\mathbf{x}) = P^2\mathbf{x} = P\mathbf{x} = \mathbf{w}$$

Since $\mathbf{v} - \mathbf{w} \in \mathrm{Null}(P)$:

$$P(\mathbf{v} - \mathbf{w}) = \mathbf{0} \quad \Longrightarrow \quad P\mathbf{v} = P\mathbf{w}$$

Combining:

$$P\mathbf{v} = P\mathbf{w} = \mathbf{w} \qquad \square$$

Summary: The Sunlight Model

Projection Operator Summary

As long as $P^2 = P$, the operator P is a projection:

1. For any $\mathbf{v} \in V$:
   - $P\mathbf{v}$ = the shadow
   - $(I - P)\mathbf{v} = \mathbf{v} - P\mathbf{v}$ = vector pointing toward the sun
2. $\text{Null}(P)$ = **direction of sunlight**
3. $\text{Col}(P)$ = **the floor**
   - For any $\mathbf{w} \in \text{Col}(P)$: $P\mathbf{w} = \mathbf{w}$

Basic Properties of Projections

Column Space Membership Test

Proposition (Floor Membership Test)

Let P be a projection. For any $\mathbf{y} \in \mathbb{R}^n$:

$$\mathbf{y} \in \mathrm{Col}(P) \quad \Longleftrightarrow \quad P\mathbf{y} = \mathbf{y}$$

Proof:

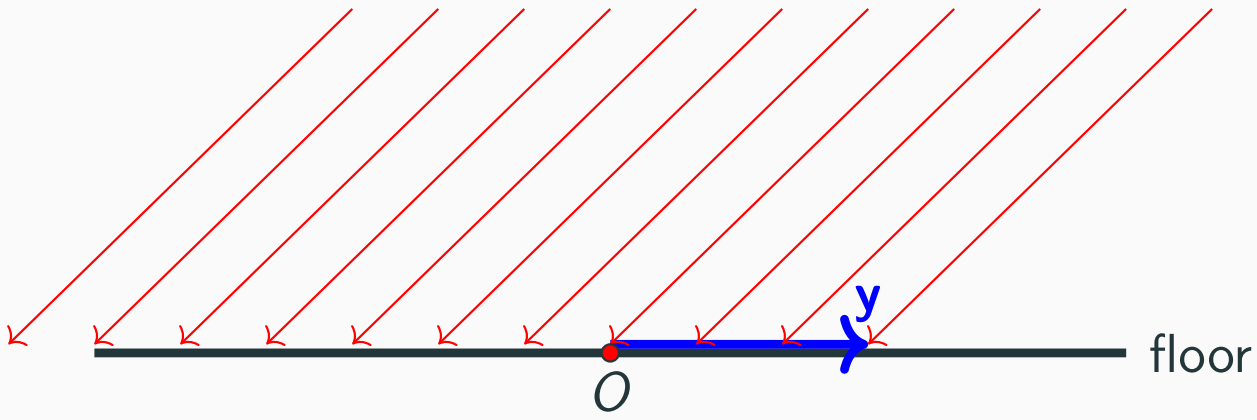
($\Rightarrow$): If $\mathbf{y} \in \mathrm{Col}(P)$, then $\mathbf{y} = P\mathbf{x}$ for some $\mathbf{x}$.

$$P\mathbf{y} = P(P\mathbf{x}) = P^2\mathbf{x} = P\mathbf{x} = \mathbf{y} \quad \checkmark$$

($\Leftarrow$): If $P\mathbf{y} = \mathbf{y}$, then $\mathbf{y} \in \mathrm{Col}(P)$ by definition. $\checkmark$

Geometric Meaning: On the Floor $\iff$ Own Shadow

Geometric meaning: A vector is on the floor $\iff$ it equals its own shadow.



$P\mathbf{y} = \mathbf{y}$: the shadow of $\mathbf{y}$ is $\mathbf{y}$ itself

A vector $\mathbf{y}$ lies on the floor means its shadow coincides with itself:

$$\mathbf{y} \in \mathrm{Col}(P) \quad \iff \quad P\mathbf{y} = \mathbf{y}$$

Exercise: Testing Membership

Exercise

The following matrix is a projection:

$$P = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

Calculate $P \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$. Is $\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \in \mathrm{Col}(P)$?

Hint: Use the membership test — check whether $P\mathbf{y} = \mathbf{y}$.

The Companion Projection: $I - P$

Proposition ($I - P$ is also a Projection)

If $P^2 = P$, then $I - P$ is also a projection.

Proof:

$$(I - P)^2 = I - 2P + P^2 = I - 2P + P = I - P \quad \checkmark$$

Geometric interpretation:

P	projects **onto the floor**
$I - P$	projects **in the direction of sunlight**

For any $\mathbf{v}$:

$$\underbrace{P\mathbf{v}}_{\text{shadow on floor}} + \underbrace{(I - P)\mathbf{v}}_{\text{displacement toward sun}} = \mathbf{v}$$

The Interchanging Property

Proposition (Column Space and Null Space Interchange)

For any projection P:

$$\mathrm{Col}(P) = \mathrm{Null}(I - P)$$

$$\mathrm{Col}(I - P) = \mathrm{Null}(P)$$

Proof (first equality):

($\subseteq$): $\mathbf{v} \in \mathrm{Col}(P) \implies P\mathbf{v} = \mathbf{v} \implies (I - P)\mathbf{v} = \mathbf{0} \implies \mathbf{v} \in \mathrm{Null}(I - P)$

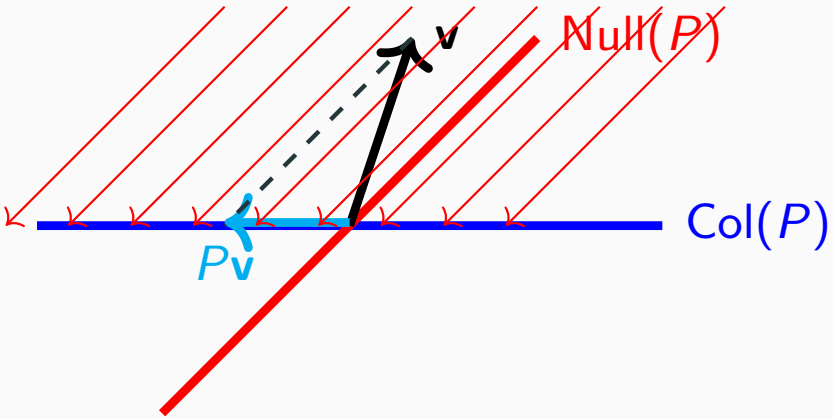
($\supseteq$): $\mathbf{v} \in \mathrm{Null}(I - P) \implies (I - P)\mathbf{v} = \mathbf{0} \implies \mathbf{v} = P\mathbf{v} \implies \mathbf{v} \in \mathrm{Col}(P)$

For P	**For** $I - P$
$\mathrm{Col}(P)$	$= \mathrm{Null}(I - P)$
$\mathrm{Null}(P)$	$= \mathrm{Col}(I - P)$

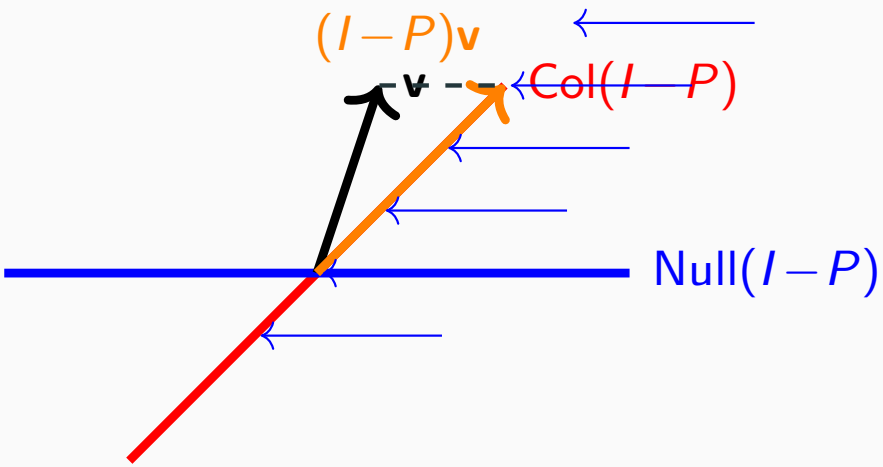
Beautiful! P and $I - P$ **interchange** their column spaces and null spaces.

Visualizing the Interchange

Projection P:



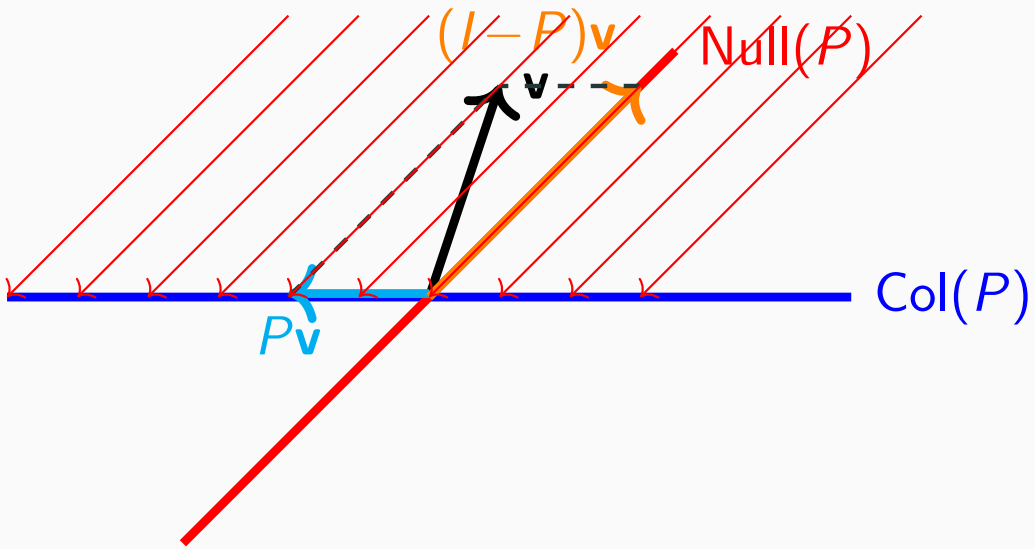
Projection $I - P$:



The roles of Col and Null are **swapped**: the floor of P becomes the sunlight of $I - P$, and vice versa.

Decomposition: $\mathbf{v} = P\mathbf{v} + (I - P)\mathbf{v}$

Every vector $\mathbf{v}$ is **decomposed** into two components:



$$\mathbf{v} = \underbrace{P\mathbf{v}}_{\in \mathrm{Col}(P)} + \underbrace{(I - P)\mathbf{v}}_{\in \mathrm{Null}(P)}$$

P and $I - P$ together **decompose** every vector into a floor component and a sunlight component.

The Disjoint Property

Proposition (Column Space and Null Space Are Disjoint)

For any projection P:

$$\mathrm{Col}(P) \cap \mathrm{Null}(P) = \{\mathbf{0}\}$$

Proof:

Let $\mathbf{v} \in \mathrm{Col}(P) \cap \mathrm{Null}(P)$.

$\mathbf{v} \in \mathrm{Col}(P) \implies P\mathbf{v} = \mathbf{v}$

$\mathbf{v} \in \mathrm{Null}(P) \implies P\mathbf{v} = \mathbf{0}$

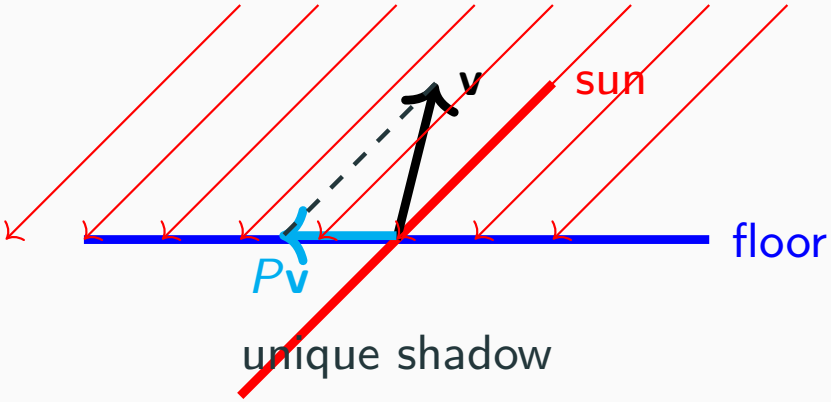
Therefore $\mathbf{v} = \mathbf{0}$. $\square$

Geometric meaning: The floor and the sunlight direction **only meet at the origin**. A non-zero vector cannot simultaneously be on the floor and point toward the sun.

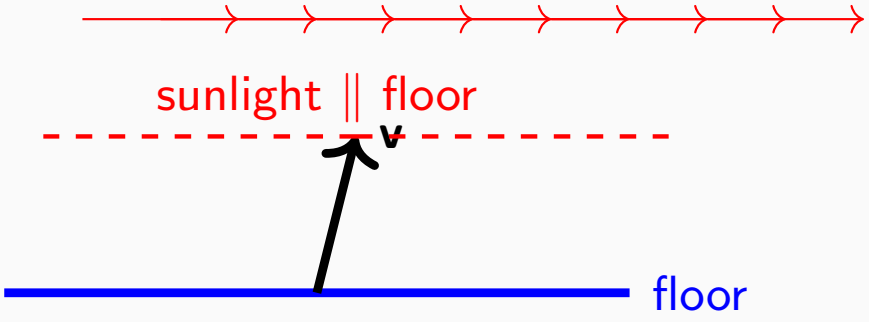
Why Disjoint? The Unique Shadow Principle

The disjoint property ensures sunlight and floor **intersect only at the origin**, so every vector has a **unique shadow**.

Good: floor $\cap$ sunlight $= \{\mathbf{0}\}$



Bad: sunlight **parallel** to floor



If sunlight is parallel to the floor, the shadow **never reaches the ground**. This cannot happen for a projection because the disjoint property guarantees a unique landing point.

Orthogonal Projections

Motivation: Perpendicular Sunlight

In Lecture 6, we discovered:

$$\mathrm{Col}(A^T) \perp \mathrm{Null}(A)$$

A natural question in the sunlight-floor model:

Can the sunlight shine perpendicular to the floor?

That is: can we have $\mathrm{Col}(P) \perp \mathrm{Null}(P)$?

Definition

Definition (Orthogonal Projection)

A projection P ($P^2 = P$) is called an **orthogonal projection** if

$$\mathrm{Col}(P) \perp \mathrm{Null}(P)$$

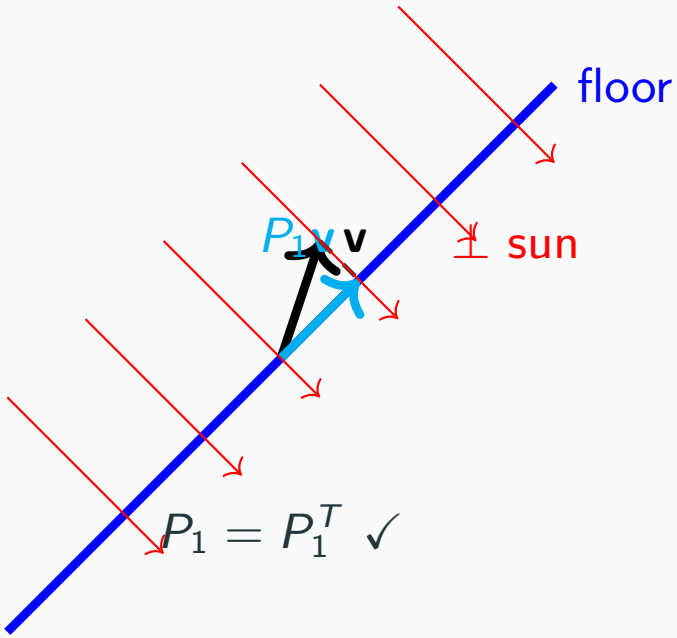
That is, $\mathbf{v}^T\mathbf{w} = 0$ for every $\mathbf{v} \in \mathrm{Col}(P)$ and $\mathbf{w} \in \mathrm{Null}(P)$.

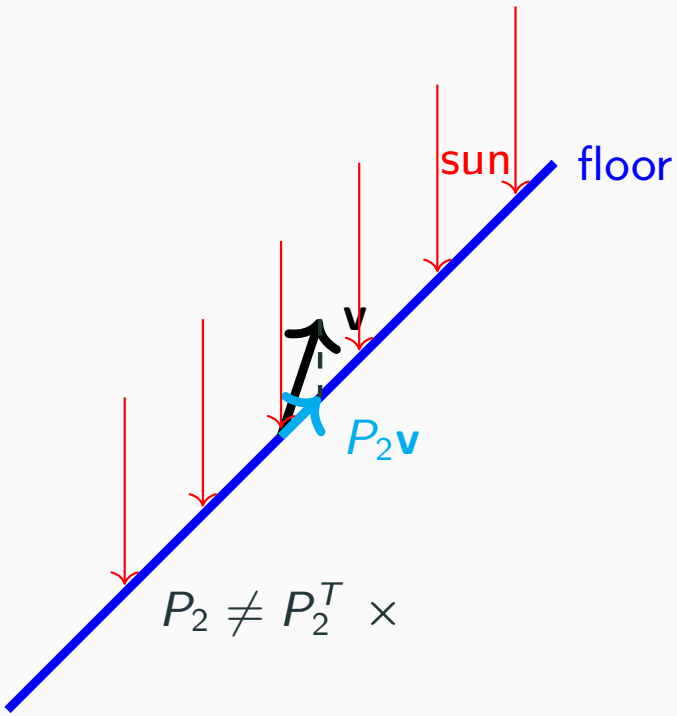
In the sunlight model: the sunlight shines **perpendicular** to the floor.

Example: Orthogonal vs Oblique

Two projections onto the **same floor** $W = \text{span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}$:

Orthogonal: $P_1 = \frac{1}{2}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}$ **Oblique**: $P_2 = \begin{pmatrix}1 & 0\\1 & 0\end{pmatrix}$





Same floor, different sunlight. Only P_1 has **perpendicular sunlight**.

Verification of the Example

Orthogonal $P_1 = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$:

- $\mathrm{Col}(P_1) = \mathrm{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}, \quad \mathrm{Null}(P_1) = \mathrm{span}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}$
- $\begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 - 1 = 0$ ✓ **Perpendicular!**
- $P_1 = P_1^T$ ✓

Oblique $P_2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$:

- $\mathrm{Col}(P_2) = \mathrm{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}, \quad \mathrm{Null}(P_2) = \mathrm{span}\left\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$
- $\begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0 + 1 = 1 \neq 0$ × **Not perpendicular!**
- $P_2 \neq P_2^T$ ×

The Characterization Theorem

Theorem (Orthogonal Projection $\Leftrightarrow$ Symmetry)

Let P be a projection ($P^2 = P$). The following are equivalent:

1. **Geometric**: $\text{Col}(P) \perp \text{Null}(P)$
2. **Algebraic**: $P = P^T$

Why this is powerful: To check orthogonality, just compare entries: $p_{ij} = p_{ji}$.

No need to compute column spaces, null spaces, or inner products!

Proof: $(P = P^T) \Rightarrow (\mathrm{Col}(P) \perp \mathrm{Null}(P))$

Assume $P = P^T$. Take $\mathbf{v} \in \mathrm{Col}(P)$ and $\mathbf{w} \in \mathrm{Null}(P)$.

By Lecture 6: $\mathrm{Col}(A^T) \perp \mathrm{Null}(A)$ always holds.

Since $P^T = P$:

$$\mathrm{Col}(P) = \mathrm{Col}(P^T) \perp \mathrm{Null}(P) \quad \checkmark$$

Alternative direct proof: Since $P\mathbf{v} = \mathbf{v}$:

$$\mathbf{v}^T\mathbf{w} = (P\mathbf{v})^T\mathbf{w} = \mathbf{v}^T \underbrace{P^T}_{=P} \mathbf{w} = \mathbf{v}^T \underbrace{P\mathbf{w}}_{=\mathbf{0}} = 0 \quad \checkmark$$

Proof: $(\mathrm{Col}(P) \perp \mathrm{Null}(P)) \Rightarrow (P = P^T)$

Assume $\mathrm{Col}(P) \perp \mathrm{Null}(P)$. Take **any** vectors $\mathbf{v}, \mathbf{w}$.

- $P\mathbf{v} \in \mathrm{Col}(P)$ (output of P is always in $\mathrm{Col}(P)$)
- $(I - P)\mathbf{w} \in \mathrm{Null}(P)$ (by interchanging property: $\mathrm{Col}(I - P) = \mathrm{Null}(P)$)

By orthogonality, $(P\mathbf{v})^T(I - P)\mathbf{w} = 0$ for all $\mathbf{v}, \mathbf{w}$.

Now expand using the **transpose undressing rule** $(AB)^T = B^T A^T$:

$$\begin{aligned} 0 &= (P\mathbf{v})^T(I - P)\mathbf{w} && \text{(orthogonality)} \\ &= \mathbf{v}^T P^T (I - P)\mathbf{w} && \text{(undressing: } (P\mathbf{v})^T = \mathbf{v}^T P^T\text{)} \\ &= \mathbf{v}^T (P^T - P^T P)\mathbf{w} && \text{(distribute)} \end{aligned}$$

Since this holds for **all** $\mathbf{v}, \mathbf{w}$: $P^T - P^T P = 0$, so $P^T = P^T P$.

Proof (continued): Concluding $P = P^T$

We showed: $P^T = P^T P$. Now take the **transpose of both sides**:

$$
\begin{aligned}
(P^T)^T &= (P^T P)^T && \text{(transpose both sides)} \\
P &= P^T \underbrace{(P^T)^T}_{=P} && \text{(undressing + } (A^T)^T = A\text{)} \\
P &= P^T P
\end{aligned}
$$

Combining: $P^T = P^T P = P$, therefore $P = P^T$. $\square$

Key principle: If $\mathbf{v}^T A \mathbf{w} = \mathbf{v}^T B \mathbf{w}$ for all $\mathbf{v}, \mathbf{w}$, then $A = B$.

Orthogonal Projection: Summary

Orthogonal Projection = Two Algebraic Conditions

$$\boxed{P^2 = P \quad \text{and} \quad P = P^T}$$

The geometry of "sunlight perpendicular to floor" is captured by **matrix symmetry**.

Exercise

Fill in the blanks so that the following matrix is a projection:

$$P = \begin{pmatrix} 0.5 & \square \\ 0.5 & \square \end{pmatrix}$$

Is your answer an orthogonal projection?

Summary

What We Learned Today

1. **Linear Transformation**: $T(\lambda \mathbf{v} + \mu \mathbf{w}) = \lambda T(\mathbf{v}) + \mu T(\mathbf{w})$
   - Represented by matrices: $T(\mathbf{v}) = A\mathbf{v}$
2. **Projection Operator**: $P^2 = P$
   - $\text{Col}(P) = \text{floor}, \quad \text{Null}(P) = \text{sunlight}$
3. **Basic Properties**:
   - $\mathbf{y} \in \text{Col}(P) \iff P\mathbf{y} = \mathbf{y}$
   - $I - P$ is also a projection
   - $\text{Col}(P) = \text{Null}(I - P)$ (interchanging)
   - $\text{Col}(P) \cap \text{Null}(P) = \{\mathbf{0}\}$ (disjoint)
4. **Orthogonal Projection**: $P^2 = P$ and $P = P^T$
   - Perpendicular sunlight $\iff$ symmetry

Preview: Lecture 8

Lecture 8: Cross-Filling Projections

What happens when we **cross-fill** a projection matrix P?

Something remarkable: each rank-one piece is **automatically a projection**!

This will lead to:

- $\text{rank}(P) = \text{trace}(P)$ — true **only** for projections
- If $UV = I$, the factors must be **square and invertible**
- Full-rank square matrices are always invertible